instituto superior de engenharia de lisboa

Electrical Engineering BSc

Electric Machinery II

Exercises and Industrial Application Problems

September 2026

Exercises and Industrial Application Problems

Electric Machinery II

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Preface

This collection of exercises and industrial application problems is organised into three chapters:

- DC machines
- Three-phase synchronous machines
- Electrical machine transients

This document presents exercises of a more conceptual nature alongside industrial application problems, allowing students to progress from fundamental concepts to practical situations encountered in industry. In each chapter, the exercises or problems are numbered sequentially (Exercise 1, Exercise 2, …), with their scope made clear by the context and problem statement itself.

The odd-numbered exercises are fully worked through, while the even-numbered exercises are proposed as work items, with only the numerical results given. The latter are primarily intended to be solved during theoretical-practical classes, adopting Problem-Based Learning as the teaching and learning methodology. In this way, discussion and critical analysis of the topics are encouraged, fostering the development of analytical thinking and problem-solving skills.

Among the worked exercises, some are solved analytically as an introduction to, or consolidation of, key concepts, while others are solved using the scientific computing language `Julia`, through computational documents (*notebooks*) developed using the integrated development environment `Pluto.jl`. The use of interactive *notebooks* allows students to deepen their understanding of fundamental theoretical concepts and to develop essential skills for professional practice. In addition, this exploratory practice allows students to apply their knowledge to solve exercises interactively, to visualise and analyse results graphically, and to gain practical experience with engineering tools.

Lisbon, September 2025

Ricardo Luís

The real process of education should be
the process of learning to think through
the application of real problems.

— **John Dewey** (1859-1952)

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DC MACHINES

Exercise 1

A six-pole DC machine has 360 conductors in armature slots. Each magnetic pole has a pole arc of 20 cm, a depth of 20 cm and a magnetic flux density of 0.8 T. With the rotor running at a speed of 1000 rpm, determine the induced electromotive force, E_0, obtained if the machine has:

a) a lap-wound armature winding;

b) a wave-wound armature winding.

Worked solution:

Let:

- $2p = 6$ poles; $z = 360$ conductors;
- $b = 20$ cm (pole arc); $l = 20$ cm (depth)
- $B = 0.8$ T

Pole face area, A_p:

$$A_p = b\ l = 0.2 \times 0.2 = 0.04 \text{ m}^2$$

The magnetic flux per pole, ϕ_p, is then:

$$\phi_p = B\ A_p = 0.8 \times 0.04 = 0.032 \text{ Wb}$$

With the speed in rpm, the armature constant, k, is given by: $k = \dfrac{z\ p}{60\ a}$

Thus, the induced electromotive force, E_0, in the armature winding is obtained:

a) Lap winding, $a = p$ (number of brushes equal to number of poles):

$$E_0 = k\phi_p n = \frac{360 \times 3}{60 \times 3} \times 0.032 \times 1000 = 192 \text{ V}$$

b) Wave winding, $a = 1$ (only one pair of brushes):

$$E_0 = k\phi_p n = \frac{360 \times 3}{60 \times 1} \times 0.032 \times 1000 = 576 \text{ V}$$

(Source: Adapted from exercise 5.5 in [Guru & Hızıroğlu, 2003])

Exercise 2

Consider the armature of a four-pole DC generator with a lap winding placed in 28 slots with 10 conductors in each. The magnetic flux per pole is 40 mWb and the armature (rotor) speed is 1200 rpm. The generator supplies a load, and the current in each conductor is 2 A. What are the torque and power developed by the generator?

> **Results:**
>
> $T_d = 14.26$ Nm ; $P_d = 1.79$ kW

(Source: Adapted from exercise 5.8 in [Guru & Hızıroğlu, 2003])

Exercise 3

Consider a DC generator with $2p = 6$ poles, a simple lap armature winding, i.e. $2a = 6$ parallel paths, and $z = 624$ active conductors uniformly distributed around the armature periphery.

The cross-sectional area of each armature conductor is $s = 8$ mm², and carries a current density of $J = 3$ A.mm^{-2}.

The machine has a pole-arc-to-pole pitch ratio, $b/\tau = 2/3$, and the armature has a diameter of $D = 220$ mm.

a) Determine the magnetomotive force (MMF) associated with the armature reaction, $\mathcal{F}_i$, along the geometric neutral axis;

b) Assuming that the machine is fitted with interpoles only, determine the number of turns to be placed on the interpoles;

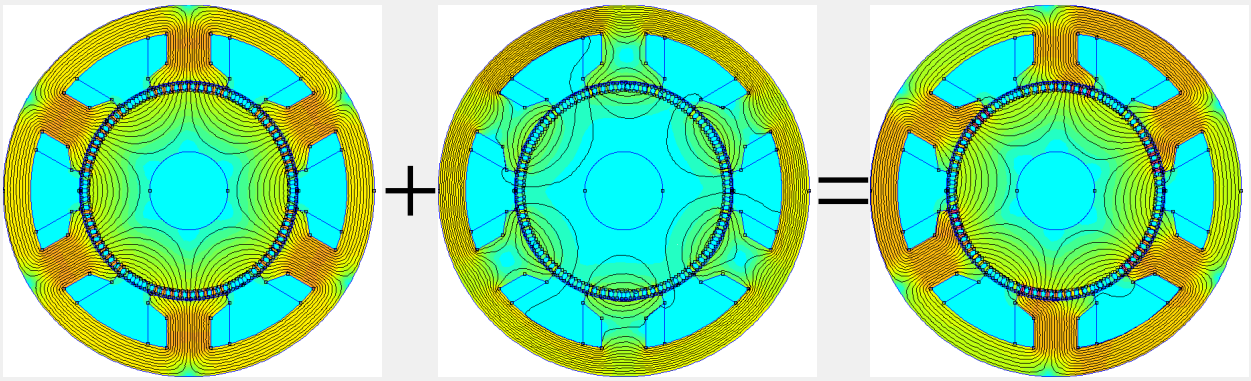
c) Suppose the the machine is to have its $\mathcal{F}_i$ fully compensated:
   1) Calculate the number of turns to be placed in a compensating winding, assuming that 6 slots were opened in each main pole. Calculate the number of conductors to be placed in each slot;
   2) Size the number of turns to be placed on the interpoles after the compensating windings have been installed.

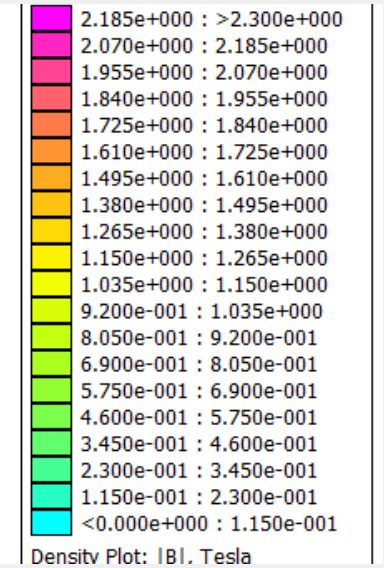
Worked solution:

Two solution approaches were used in this exercise:

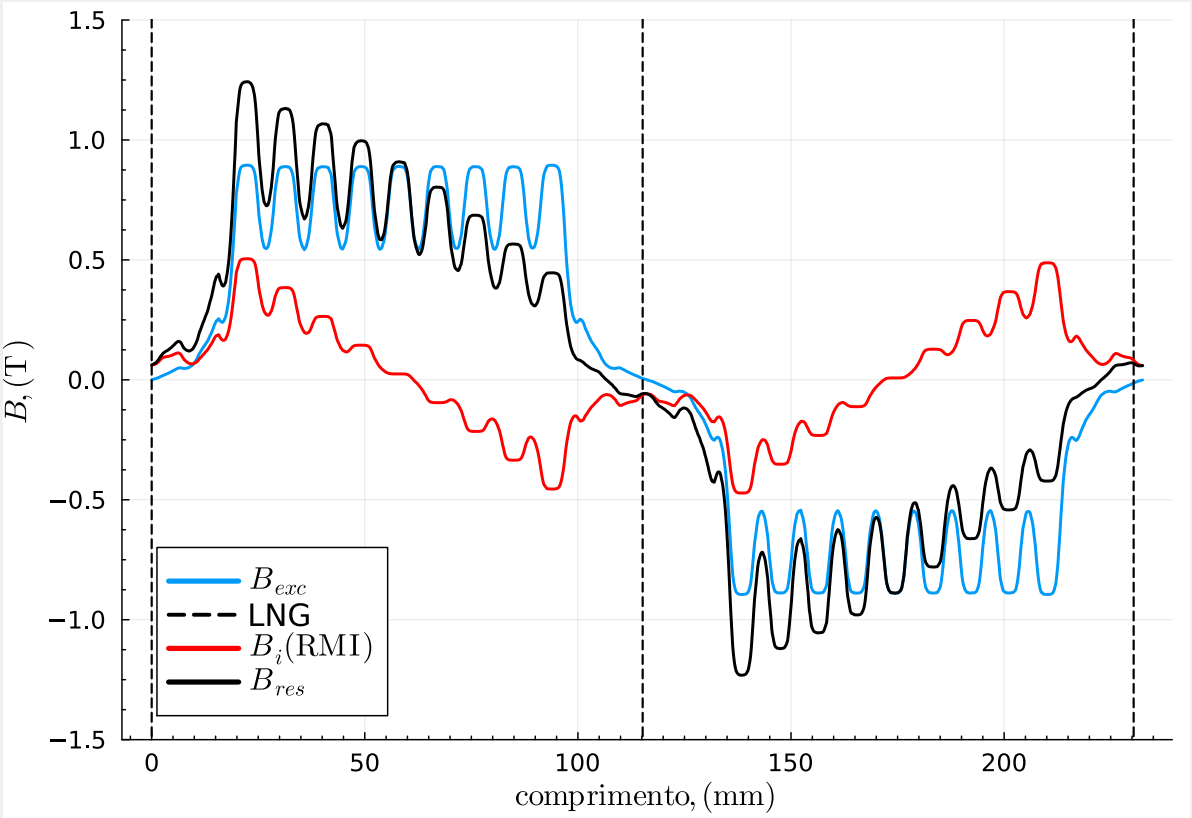
- **numerical (computational) solution**: using the exercise data, a CAD (*Computer-Aided Design*) and CAE (*Computer-Aided Engineering*) model was built, applying the finite element method for the electromagnetic calculation. The *software* used was FEMM - Finite Element Method Magnetics. The computational solutions make it possible to visualise and quantify the magnetic flux lines, helping to understand the armature magnetic reaction and its effects on the machine's operation;
- **Analytical solution**: this is the main objective of this exercise, namely to compensate the Armature Magnetic Reaction (AMR), by calculating the magnetomotive forces involved.

field winding + armature field = resultant field

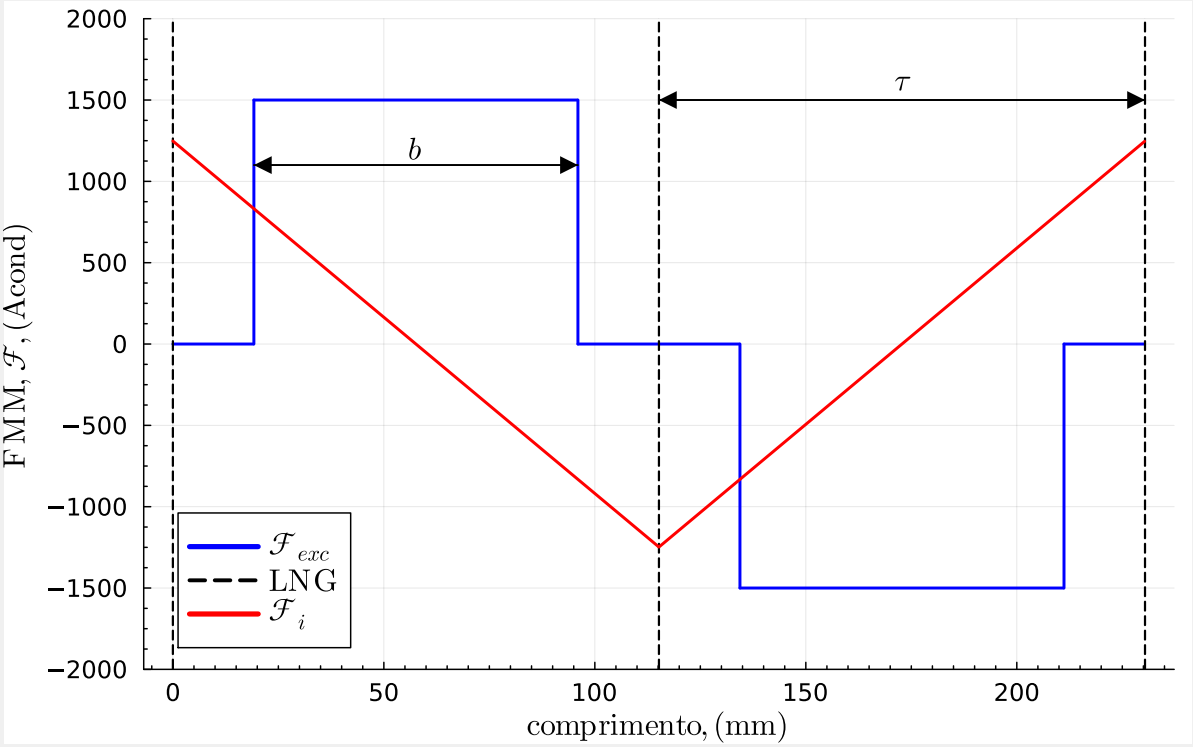




field winding + armature field = resultant field



MMF of the field winding and the armature reaction



a)

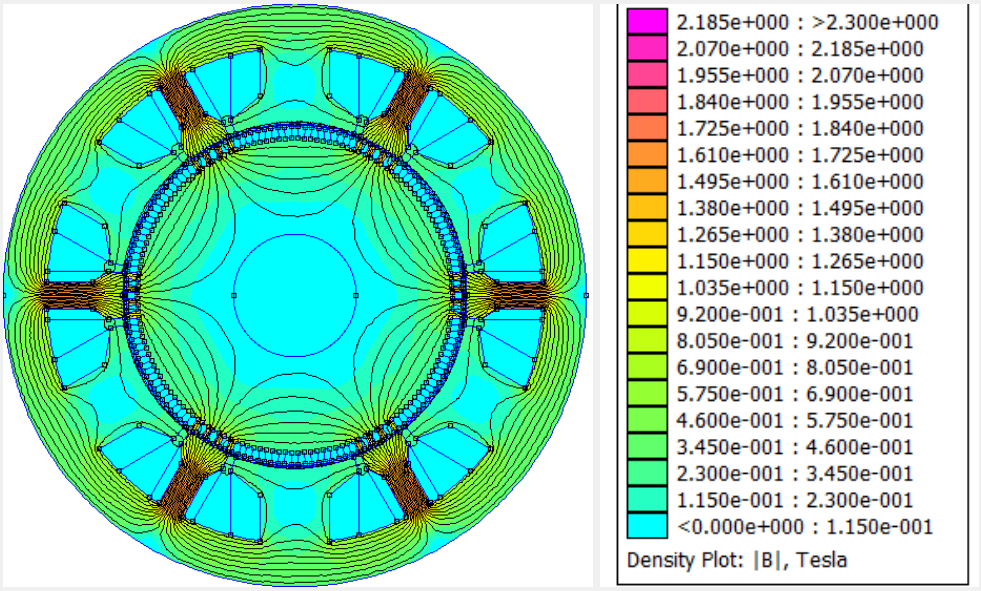
$I_a = J\ s = 3 \times 8 = 24\ \text{A}$

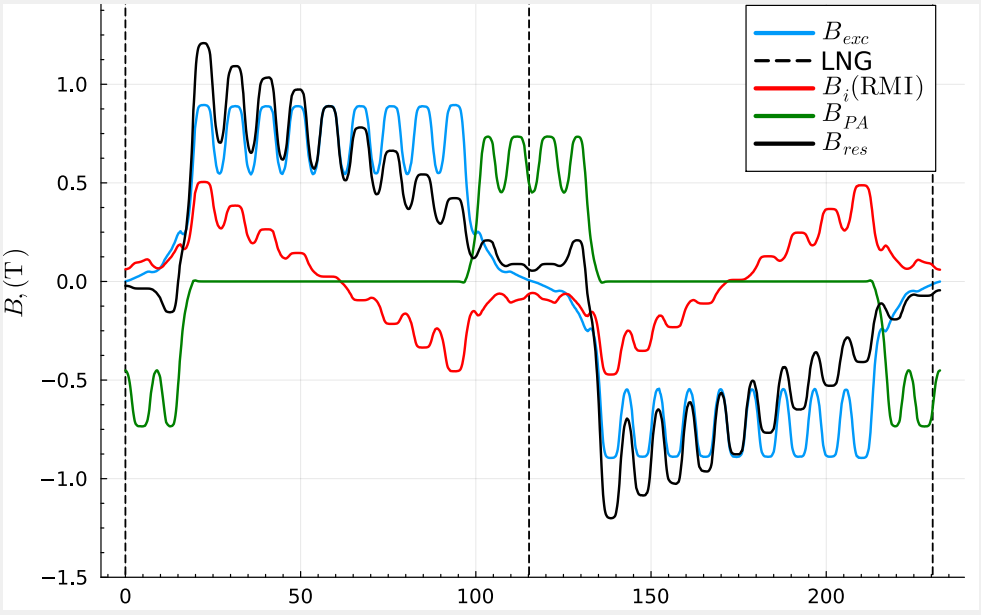
$I = I_a\ 2a = 24 \times 2 \times 3 = 144\ \text{A}$

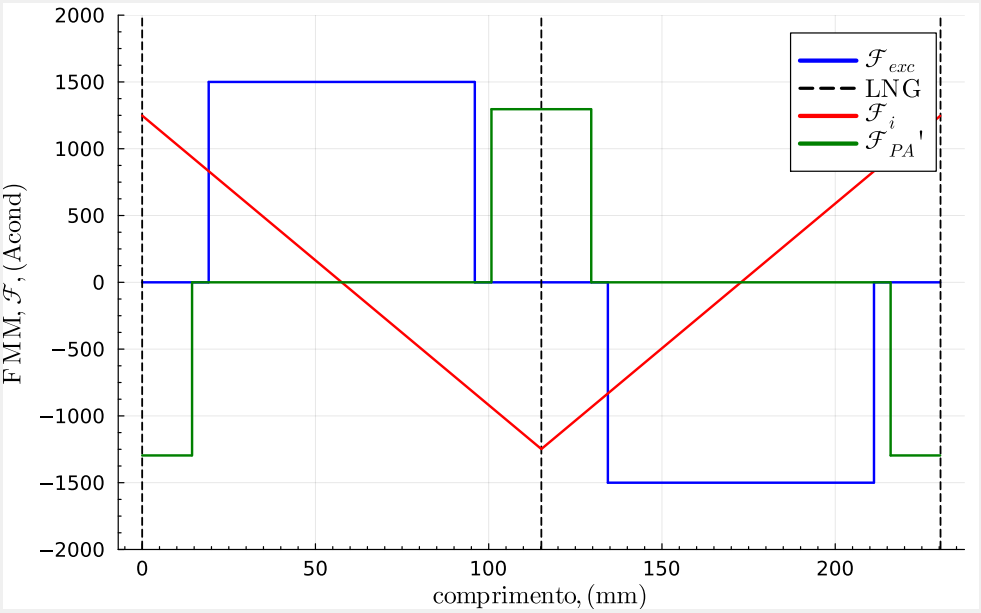
$\mathcal{F}_i = A\ \frac{\tau}{2}$ with: $A = \frac{z\ I}{4\ a\ p}\ \frac{1}{\tau}$ or $\mathcal{F}_i = \frac{z\ I}{8\ a\ p}$, using this latter expression:

$\mathcal{F}_i = \frac{z\ I}{8\ a\ p} = \frac{624 \times 144}{8 \times 3 \times 3} = 1248\ \text{A.cond}$

b) DC machine with **interpoles**:







For a machine with interpoles:

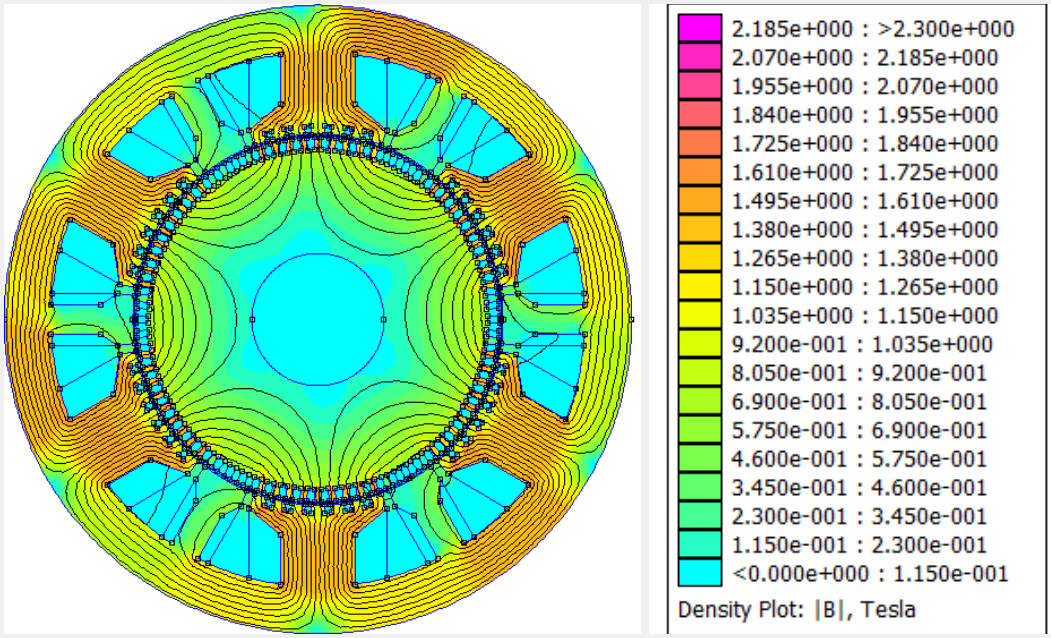
$\mathcal{F}_{\mathrm{PA}} = \mathcal{F}_{\mathrm{i}}$ since: $\mathcal{F}_{\mathrm{PA}} = N_{\mathrm{PA}}\ I$, we obtain: $N_{\mathrm{PA}} = \dfrac{\mathcal{F}_{\mathrm{i}}}{I} = \dfrac{1248}{144} = 8.67$ turns

To facilitate commutation, the next integer above is used, i.e. 9 turns.

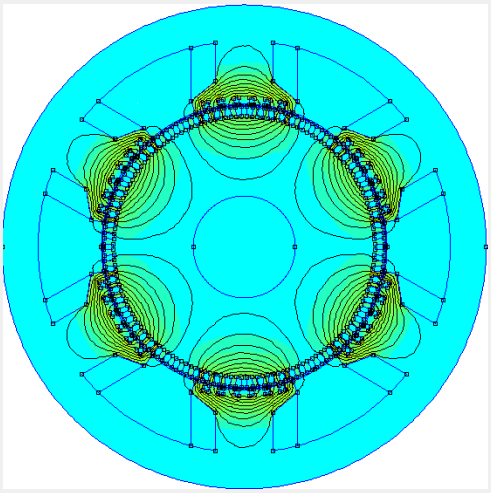
Thus, a corrected MMF is obtained for the interpoles, $\mathcal{F}'_{\mathrm{PA}}$:

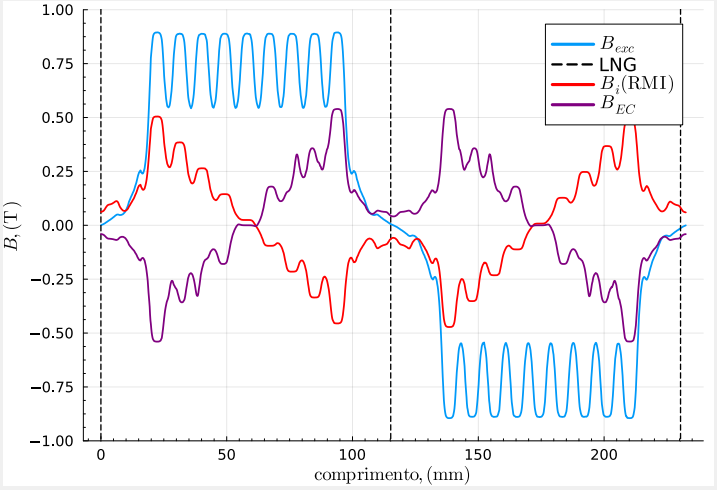
$\mathcal{F}'_{\mathrm{PA}} = 9 \times 144 = 1296$ A.cond

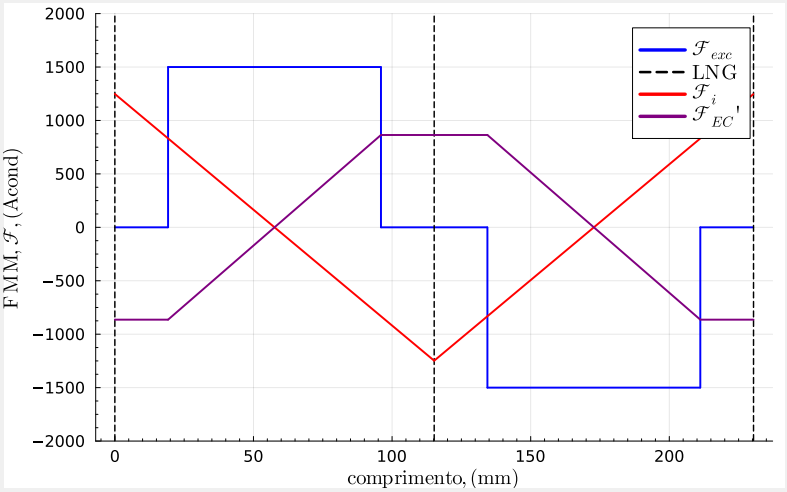
c) Machine with compensated AMR (**interpoles and compensating windings**)



c-1) The **compensating windings** are sized first:







$\mathcal{F}_{\mathrm{EC}} = A\,\dfrac{b}{2}$ or $\mathcal{F}_{\mathrm{EC}} = \mathcal{F}_{\mathrm{i}}\,\dfrac{b}{\tau}$, using this latter expression since $\dfrac{b}{\tau}$ is known:

$$\mathcal{F}_{\mathrm{EC}} = \mathcal{F}_{\mathrm{i}}\,\frac{b}{\tau} = 1248 \times \frac{2}{3} = 832 \text{ A.cond}$$

On the other hand, since $\mathcal{F}_{\mathrm{EC}} = N_{\mathrm{EC}}\ I$, the number of turns of the compensating winding is obtained as:

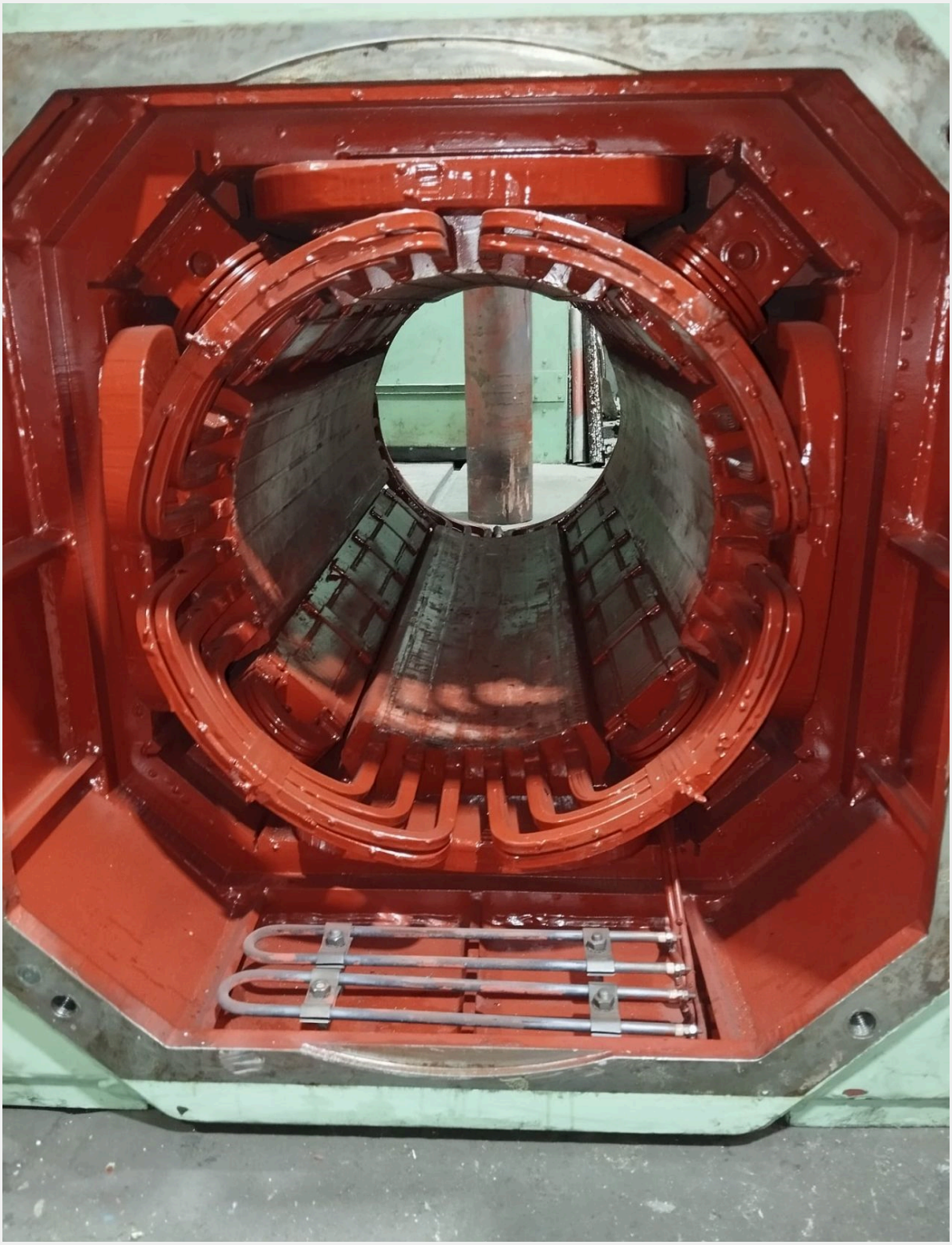
Thus, $\quad N_{\mathrm{EC}} = \dfrac{\mathcal{F}_{\mathrm{EC}}}{I} = \dfrac{832}{144} = 5.78 \text{ turns}$

Considering 6 slots per main pole, where the turns from 3 slots connect to the following pole and those from the other 3 slots connect to the preceding pole, $N'_{\mathrm{EC}} = 6$ turns are obtained, placing 2 turns in each slot.

Thus, a corrected MMF is obtained for the compensating windings, $\mathcal{F}'_{\mathrm{EC}}$:

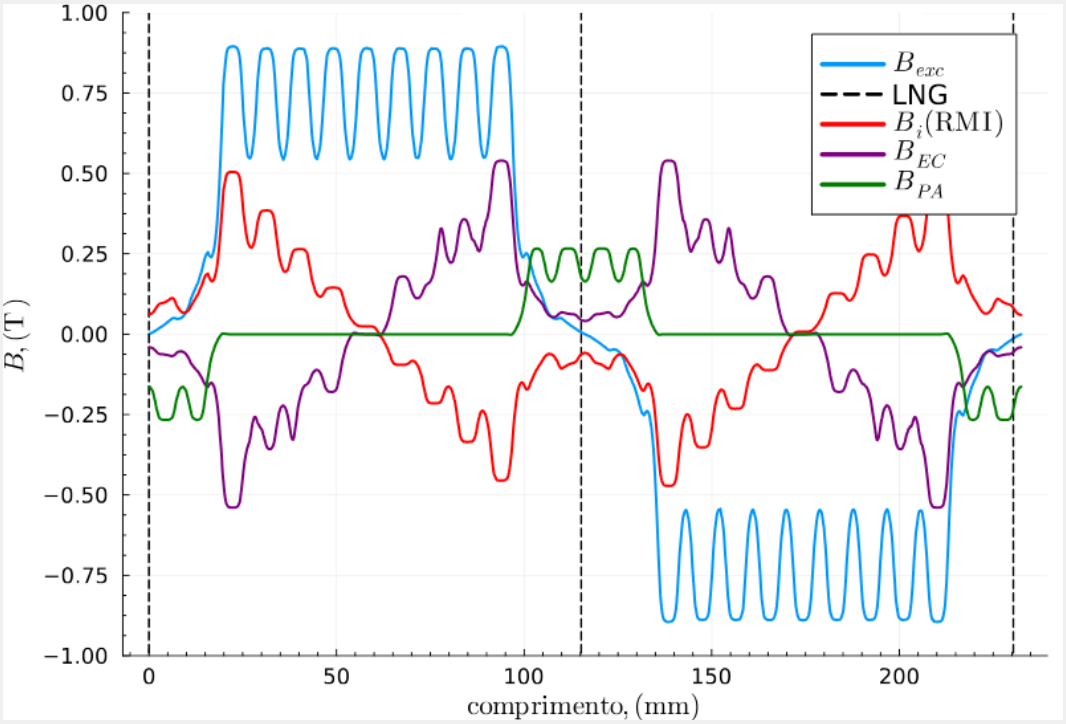
$$\mathcal{F}'_{\mathrm{EC}} = 6 \times 144 = 864 \text{ A.cond}$$

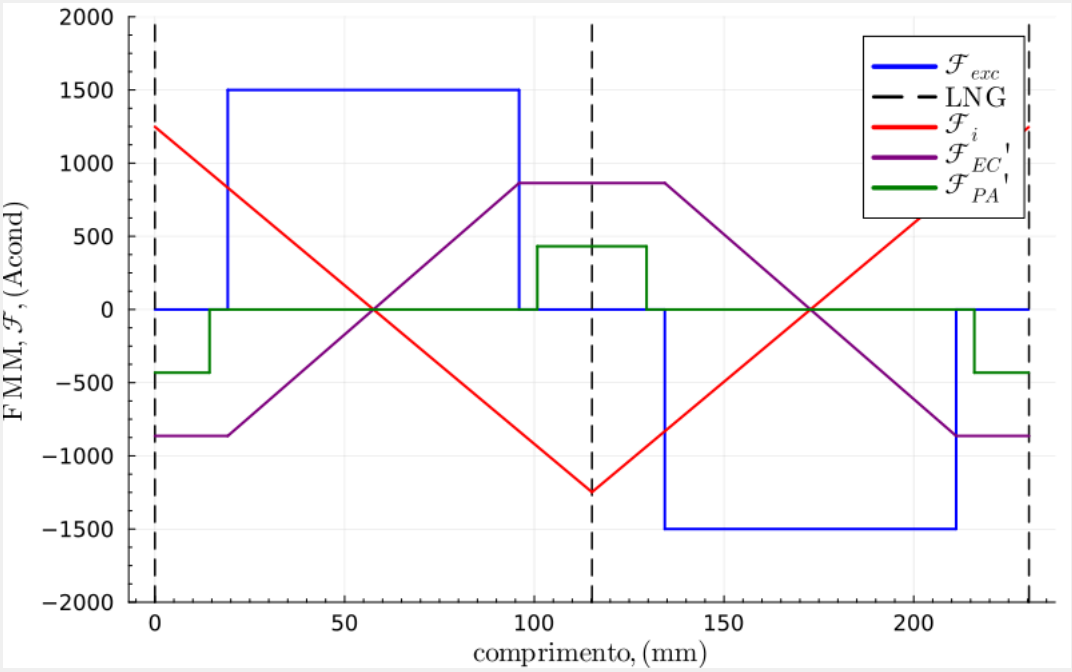
Example photograph of a machine with compensating windings (and interpoles) with 6 slots in the pole-shoe head and 2 turns per slot:



Photograph from refurbishment work on a DC machine.
Image of the stator with interpoles and compensating windings (source: Lusonave).

c-2) and then the **interpoles**:





$$\mathcal{F}_{\mathrm{PA}} = \mathcal{F}_{\mathrm{i}} - \mathcal{F}'_{\mathrm{EC}} = 1248 - 864 = 384 \text{ A.cond}$$

$$N_{\mathrm{PA}} = \frac{\mathcal{F}_{\mathrm{PA}}}{I} = \frac{384}{144} = 2.67 \text{ cond}$$

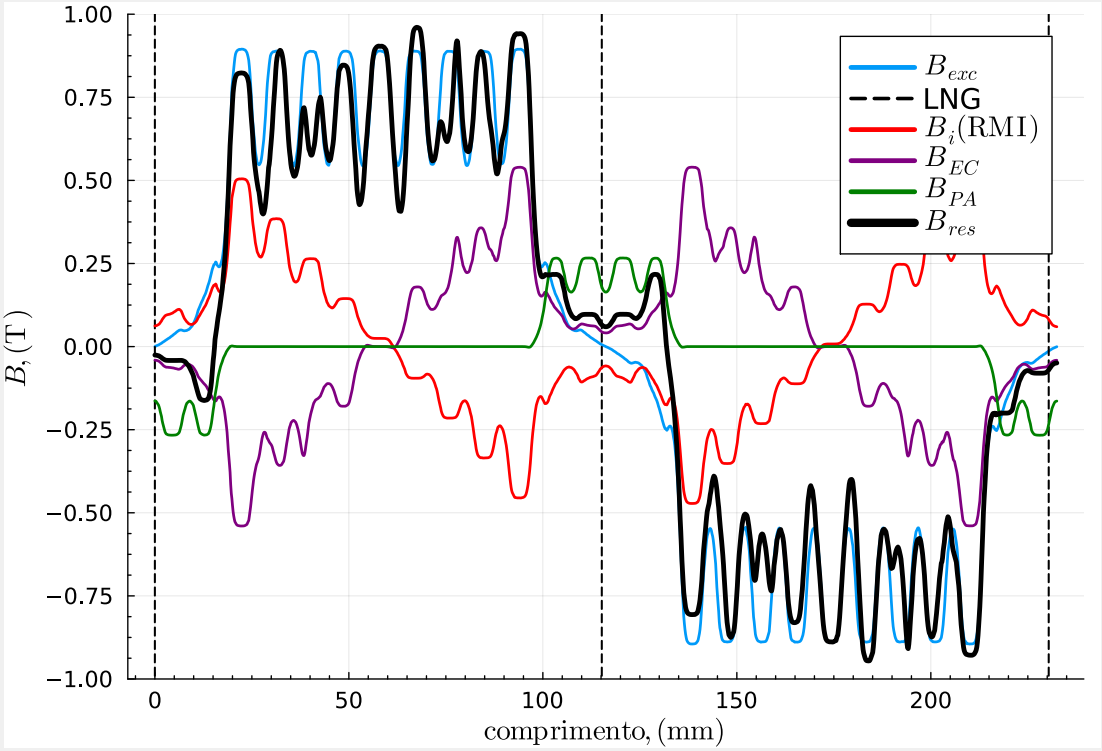
To facilitate commutation, the next integer above is used, i.e. 3 turns.

Thus, a corrected MMF is obtained for the interpoles, $\mathcal{F}'_{\mathrm{PA}}$:

$\mathcal{F}'_{\mathrm{PA}} = 3 \times 144 = 432$ A.cond

Conclusion: after fitting the compensating windings and the interpoles, a resultant MMF, $\mathcal{F}_R$, is obtained at the Geometric Neutral Lines (brush positions):

$\mathcal{F}_R = \mathcal{F}_i - \mathcal{F}'_{\mathrm{PEC}} - \mathcal{F}'_{\mathrm{PA}} = 1248 - 864 - 432 = -48$ A.cond



Exercise 4

An armature winding of a dynamo has $z = 640$ conductors, wound as a simple lap winding, for a rated current of 100 A and 2 pole pairs.

a) Size the number of turns of the compensating windings, N_{EC}, and of the interpoles, N_{PA}, for full compensation of the armature magnetic reaction. Consider that the machine's field poles cover 70 % of the total periphery, as seen from the rotor;

b) Resize N_{EC} and N_{PA} considering a simple wave armature winding.

Results:

a) $N_{\mathrm{EC}} = 14$ turns; $N_{\mathrm{PA}} = 6$ turns

b) $N_{\mathrm{EC}} = 28$ turns; $N_{\mathrm{PA}} = 12$ turns

Exercise 5

A separately-excited DC generator $[220\ \mathrm{V},\ 12\ \mathrm{A},\ 1500\ \mathrm{rpm}]$ was tested at no-load and under load, at rated speed, yielding the following characteristics:

E_0 [V]	20	180	238	270	284	300
i [A]	0.00	0.25	0.50	0.75	1.00	1.50

U [V]	278	260	242	216	186	156	128	96	62
I [A]	0	5	10	15	20	25	30	35	40

a) Determine the total internal voltage drop of this generator;

b) What is the critical resistance of a shunt-excited generator? What is its significance? How is it determined (approximately) in practice?

c) What is the field winding resistance, given that, when operating as a shunt generator, at rated speed, with no field rheostat, $U_0 = 294$ V;

d) Explain qualitatively the influence that varying the field rheostat has on the external characteristic of the shunt generator. Justify briefly;

e) Under the excitation conditions of part c), what should be done to obtain a no-load voltage of 336 V?

Worked solution to Exercise 5: 🔗 interactive notebook developed with **Pluto**.jl – julia.

Exercise 6

A shunt-excited generator has the following no-load (magnetisation) characteristic, at 1500 rpm:

i[A]	0	0.1	0.2	0.3	0.4	0.6	0.8	1.0
E[V]	20	120	200	235	250	270	285	300

$R_a = 2.0\ \Omega$

$R_d = 300\ \Omega$

Consider that the generator is fitted with interpoles and compensating windings.

a) With the generator driven at 1500 rpm, determine the value of the field rheostat to obtain the operating point $(35\ \text{A}; 180\ \text{V})$;

b) Determine the generator drive speed required for the machine operate under the same conditions as in part a), without a field rheostat;

c) Under the conditions of part a) (1500 rpm and field rheostat), determine the equivalence ratio (N_s/N_d), to obtain the operating point $(35\ \text{A};\ 200\ \text{V})$. Consider that the series field winding has a resistance of $0.5\ \Omega$.

Results:

a) $150\ \Omega$; **b)** 1389 rpm; **c)** $0.39/35$

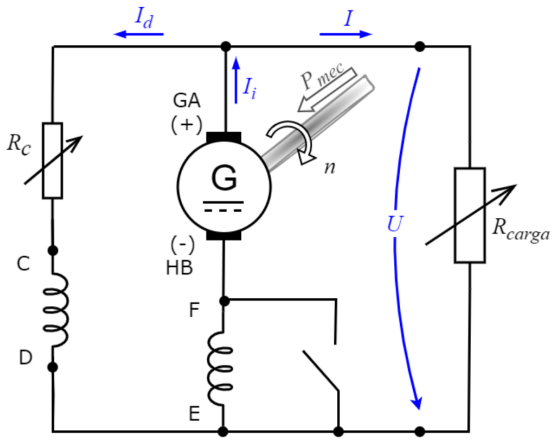
Exercise 7

A long-shunt compound generator, with cumulative (additive) series field flux, has the following characteristics:

$P = 8.75$ kW	$U = 250$ V	$R_d = 223\ \Omega$	$R_a = 1.3\ \Omega$	$R_s = 0.1\ \Omega$
$n = 1500$ rpm	$n_{\text{mag}} = 1500$ rpm	$N_d = 2000$ turns	$N_s = 50$ turns	$E_0 = \text{f}(I_{\text{exc}})$:

I_{exc} [A]	0.0	0.2	0.4	0.6	0.8	0.9	1.0	1.1	1.3	1.5	1.7
E_0 [V]	15	100	170	220	250	260	270	280	290	300	310

Also consider a switch connected in parallel with the series field winding, as show in the following electric diagram:



a) Calculate the no-load voltage with no field rheostat. Explain whether the state of the switch influences the generator no-load voltage;

b) With the switch open, calculate the voltage drop due to the armature reaction, knowing that flat (level) regulation is obtained under rated conditions;

c) With the switch closed, explain qualitatively the generator external characteristic. What is the variation of the operating point on the external characteristic, for a given load resistance, in the following situations:

1) an increase in drive speed;

2) a decrease in the shunt-field rheostat.

Worked solution to Exercise 7: 🔗 interactive notebook developed with **Pluto**.jl – julia.

Exercise 8

Consider a series generator with 5 turns/pole, whose magnetisation characteristic passes through the following points, at 1200 rpm:

$\mathcal{F}$ [Ae]	0	200	600	1000
E_0 [V]	5	120	160	170

It is known that the rotational losses are 1200 W, $R_a = 0.2\ \Omega$ and $R_s = 0.05\ \Omega$.

a) Determine the voltage drop due to the armature reaction, when supplying a $1\ \Omega$ load, at 120 A;

b) Consider that the armature reaction results in a flux loss of $5\ \%$. Will it be possible to obtain a voltage of 150 V, at the same current? Justify;

c) An attempt was made to operate this series dynamo on a $3\ \Omega$ load, but this was not possible. Comment on the situation;

d) Determine the current value corresponding to maximum efficiency.

Results:

a) 10 V ; **d)** 69.3 A

Exercise 9

The external characteristics of two shunt-excited dynamos, rated at 220 V, 110 kW, and connected in parallel, are known:

I [A]	0	200	400	500	700	900
U_1 [V]	229.5	226.5	222.5	220.0	213.0	205.5
U_2 [V]	224.0	223.0	221.0	220.0	217.5	214.0

a) How would the two machines share a current of 1500 A? What is the voltage?

b) What happens for small loads, close to zero? $0 < I < 250$ A

c) Complete:

"Under overload, the machine with _____ regulation supplies _____ current".

Worked solution to Exercise 9: 🔗 interactive notebook developed with **Pluto**.jl – julia.

Exercise 10

Two identical shunt-excited dynamos have the following magnetisation characteristic at 1500 rpm:

i [A]	0.0	0.2	0.6	1.0	1.4
E [V]	20	200	220	240	260

The armature resistance is $0.4\ \Omega$ and the field winding resistance is $150\ \Omega$; each also has a field rheostat connected, with a maximum value of $300\ \Omega$.

The two machines are connected in parallel, supplying a load $(100\ \text{A},\ 200\ \text{V})$, with **dynamo A** running at 1500 rpm.

a) Initially **dynamo B** is unloaded. Determine the field rheostat of **dynamo A**;

b) Determine the short-circuit current of **dynamo A**;

c) **Dynamo A** supplies 75 A of the total load. How should one proceed so that **dynamo B** supplies the rest? Show the calculations.

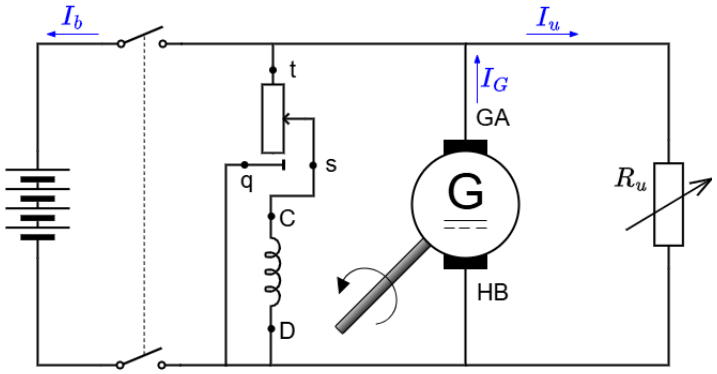
Results:

a) $R_f^{\mathrm{A}} = 50\ \Omega$; **b)** 50 A

c) Adjusting only on the field rheostat results in $R_f^{\mathrm{B}} > R_f^{\mathrm{máx}}$, so the speed must also be adjusted. For example, considering $R_f^{\mathrm{B}} = 50\ \Omega$, 1312.5 rpm is obtained.

Exercise 11

Consider a shunt generator driven at constant speed, supplying a load resistance $R_u = 7\ \Omega$. A lead-acid battery is occasionally connected to this circuit for charging, according to the following electrical circuit:



External characteristic of the generator:

I_G [A]	0.0	10	20	30	40	50	60
U_G [V]	312	300	284	262	234	200	160

The battery consists of 100 cells in series and is charged whenever its SOC (State Of Charge) reaches $25\ \%$ (approximately 1.95 V/cell). The internal resistance of each battery cell is 8 mΩ.

Determine the operating voltage of the circuit when the battery is connected (generator + load R_u), the generator current, I_G, the current in the load, I_u, and the initial battery charging current, I_b.

Worked solution:

The battery open-circuit voltage, E_b, for SOC$= 25\ \%$, is given by:

$$E_b = 100 \times 1.95 = 195\ \mathrm{V}$$

The internal resistance of the battery, R_b, with its cells in series:

$$R_b = 100 \times 8 \times 10^{-3} = 0.8\ \Omega$$

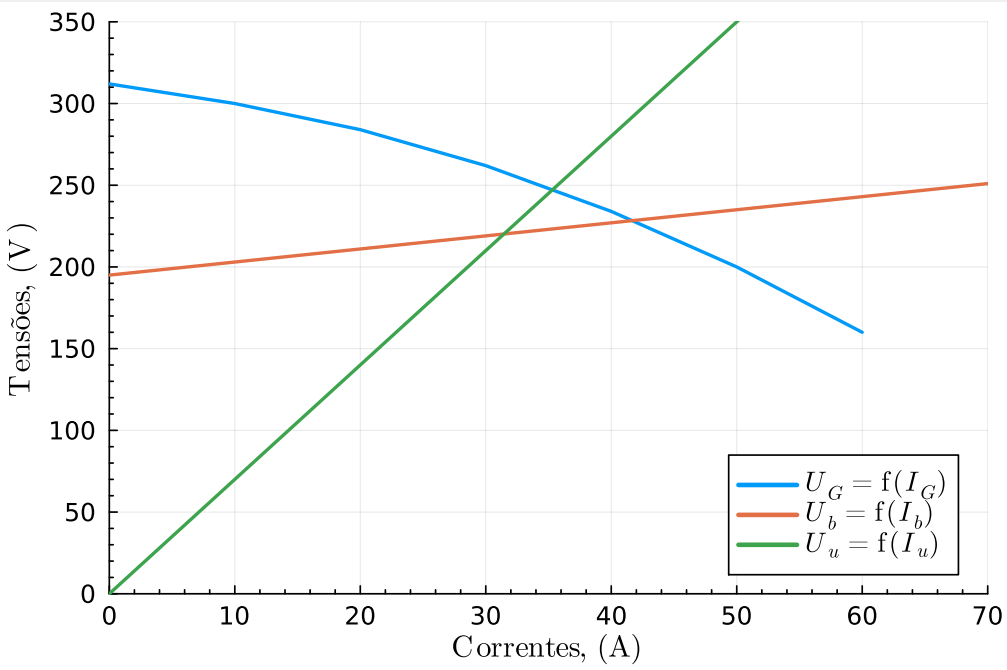
The battery charging equation, $U_b = \mathrm{f}\,(I_b)$, is given by:

$$U_b = E_b + R_b\, I_b \ \Leftrightarrow\ U_b = 195 + 0.8\, I_b$$

The following graph shows:

- the generator external characteristic, $U_G = \mathrm{f}(I_G)$
- the load-resistor line, $U_u = \mathrm{f}(I_u)$
- the battery charging line, $U_b = \mathrm{f}(I_b)$

The generator loads (the load resistor and the battery) must be combined in order to determine the operating point of the generator and of the loads.



The combined load must reflect the operation of the circuit, that is:

- the voltages across each branch are equal: $U = U_G = U_u = U_b$
- the generator supplies the load resistor and the battery: $I = I_G \approx I_u + I_b$

Note: The generator field current, I_d, is much smaller than the generator armature current and the load currents, so it is neglected in the graph:

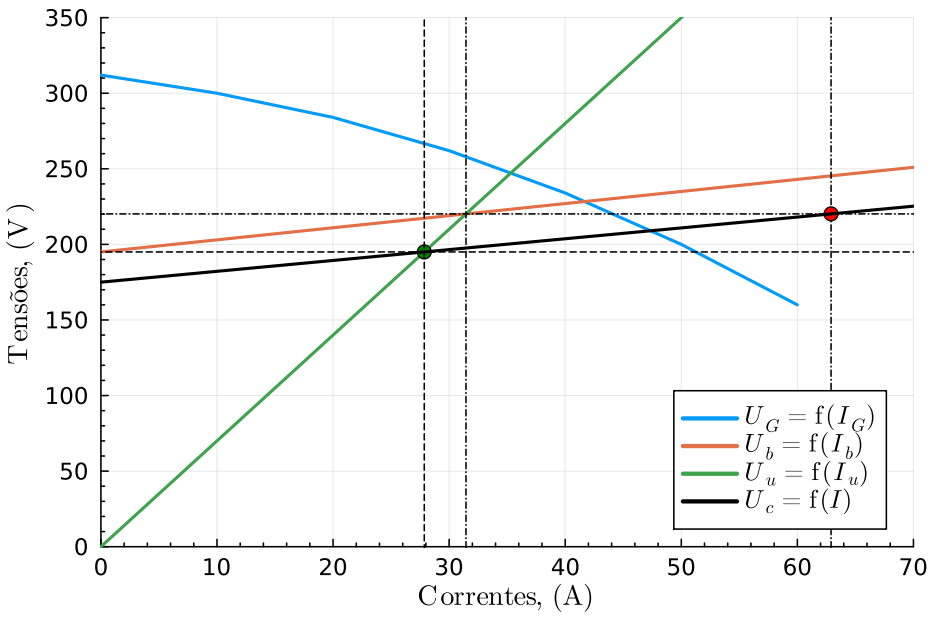
$$I_d \ll \{I_b, I_u, I_G\}$$

The following graph illustrates the graphical determination of the combined load characteristic, $U_c = \mathrm{f}\,(I)$, with $I = I_b + I_u$, corresponding to the combined loads. Since the sum of two straight lines is another straight line, only 2 voltage levels need to be considered to obtain 2 points on the combined load characteristic. Thus:

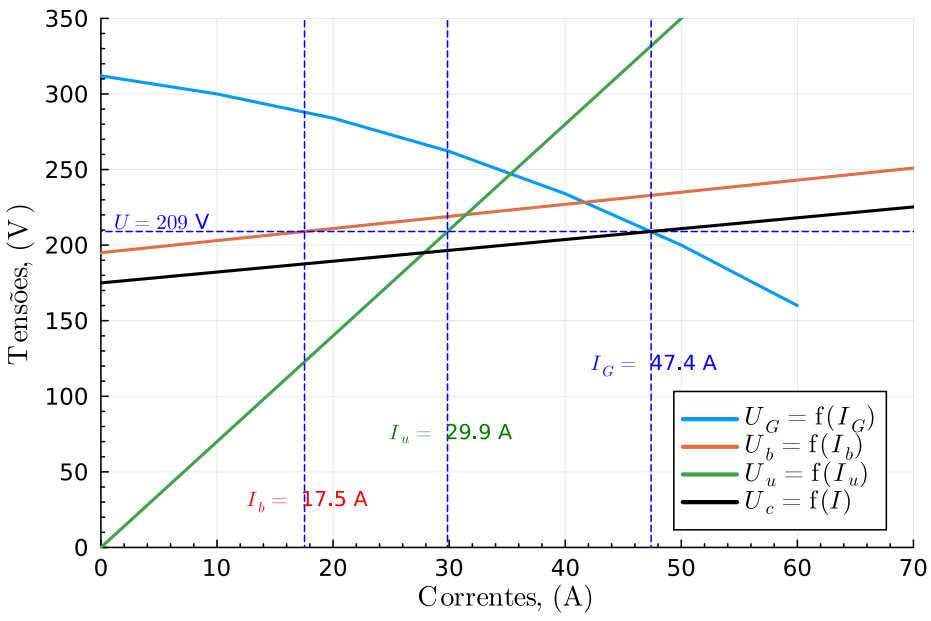
- the green point was obtained by choosing the voltage level corresponding to the battery being disconnected, i.e. E_b. In this situation, $I_b = 0$ and therefore $I = I_u$
- the red point was obtained by choosing the voltage level at which the load-resistor and battery lines intersect, i.e. $I_u = I_b$. At this voltage level, the red point is placed at the abscissa corresponding to twice the current.

Joining the points (green and red) gives the combined load characteristic:

$U_c = \mathrm{f}\,(I)$



The intersection of the combined load characteristic with the external characteristic of the DC generator defines the operating point of the circuit: $(I_G; U) = (47.4\ \mathrm{A}; 209\ \mathrm{V})$.



On the other hand, from the operating voltage, $U = 209$ V, the load lines of the load resistance and of the battery are consulted to obtain the respective current values, I_u and I_b, supplied by the generator:

$$I_u = 29.9 \text{ A} \qquad ; \qquad I_b = 17.5 \text{ A}$$

As an alternative to the graphical determination of the combined load characteristic, since it corresponds to the sum of two straight lines, it can be determined analytically. Given that:

$$U_c = U_b = U_u \quad \text{and that} \quad I = I_b + I_u \quad \text{then:}$$

$$I = \frac{U_c - E_b}{R_b} + \frac{U_c}{R_u}$$

Rearranging the previous equation the combined load line (load resistance + battery), $U_c = \text{f}\,(I)$:

$$U_c = \frac{R_u}{R_u + R_b} E_b + \frac{R_u R_b}{R_u + R_b} I$$

Exercise 12

A series motor with an armature resistance of $0.8\ \Omega$ and a field winding resistance of $0.4\ \Omega$ was run as a dynamo:

- Separately excited at 50 A and driven at 1500 rpm, it produced 230 V at the open-circuit terminals;
- With series excitation, delivering 50 A at the same speed, it produced 150 V.

Knowing that the mechanical and magnetic losses, $p_{(\text{mec}+\text{Fe})}$, are 400 W, calculate:

a) The speed when it draws 50 A as a series motor from a 220 V supply;

b) The mechanical torque it delivers to the load under these conditions;

c) Sketch qualitatively the torque and speed characteristics of this motor. Explain the effect of introducing a field rheostat on these characteristics.

Results:

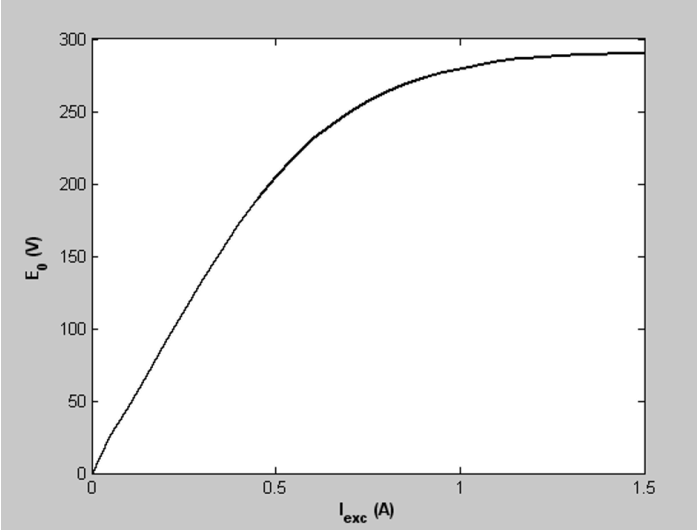
a) 1174 rpm ; **b)** 61.8 Nm

Exercise 13

Consider a DC motor with the following nameplate ratings: 17 kW, 250 V, 1200 rpm, $\eta = 85\ \%$.

The following parameters are also known:

$E_0 = f(I_{\text{exc}})$, $n = 1200$ rpm
$R_a = 0.6\ \Omega$
$R_s = 0.1\ \Omega$, 30 turns
$R_d = 200\ \Omega$, 3000 turns



a) With the motor shunt-excited, determine the value of the field rheostat under rated conditions (U_n, I_n, n_n);

b) Using the field rheostat calculated in the previous part, determine the speed, torque and mechanical characteristics of this motor (shunt excitation);

c) Repeat for cumulative and differential compound, long-shunt excitation. Plot the characteristics on the same graphs for comparison;

d) Determine the characteristic curves with the shunt circuit disconnected (series motor). Plot the characteristics on the same graphs for comparison;

e) Consider the motor with separate excitation, $U_{\text{exc}} = 240$ V, with the field rheostat calculated in part a). Explain how the speed characteristic changes in the following situations:

1) an increase in armature voltage;

2) a decrease in the field rheostat;

3) an increase in the additional resistance.

Worked solution to Exercise 13: 🔗 interactive notebook developed with **Pluto**.jl – julia.

Exercise 14

A shunt-excited motor has the following characteristics:

$$U_n = 220\ \text{V} \qquad I_n = 42\ \text{A} \qquad n_n = 1500\ \text{rpm} \qquad R_a = 0.34\ \Omega$$

a) In the no-load test the following values were obtained:

$$U = 220\ \text{V} \qquad n = 1600\ \text{rpm} \qquad I_a = 3\ \text{A} \qquad I = 4.6\ \text{A}$$

Calculate the loss torque;

b) Obtain the speed equation, $n = \text{f}\ (I_a)$;

c) Size a stepped starting rheostat such that: $40\ \text{A} \leq I_{\text{start}} \leq 65\ \text{A}$.

Results:

a) 3.92 Nm **b)** $n(\text{rpm}) = \dfrac{\text{-}100}{37.4}(I_a - 3) + 1600$

c) Resistances between the 6 contacts (Ω): 1.30; 0.80; 0.49; 0.30; 0.15. Rheostat: $3.04\ \Omega$.

Exercise 15

A series motor rated at 12 kW, 250 V, 1400 rpm, with an efficiency of $80\,\%$ and a maximum speed of 2400 rpm, has the following magnetic characteristic obtained at 1500 rpm:

I_{exc} (A)	10	20	30	40	50	60	70	80
E_0 (V)	80	140	190	225	250	270	285	295

Knowing that the armature resistance is $0.35\ \Omega$ and the field resistance is $0.1\ \Omega$, calculate:

a) The mechanical and iron losses, $p_{(\text{mec+Fe})}$;

b) The minimum current the motor can draw;

c) The voltage drop due to armature reaction at full load;

d) The motor power corresponding to maximum efficiency;

e) Explain qualitatively the effect of the field rheostat on the speed characteristic of the series motor;

f) Explain qualitatively the effect of the field rheostat on the torque characteristic of the series motor.

Worked solution to Exercise 15: 🔗 interactive notebook developed with **Pluto**.jl – julia.

Exercise 16

A series motor supplied at 250 V has an armature resistance of 0.2 Ω and a field winding resistance of 0.15 Ω. This field winding has a field rheostat of 0.1 Ω fitted with a switch.

a) Assume the switch is open. Under these conditions, the motor delivers a shaft torque of 25 Nm, running at 800 rpm and drawing a current of 10 A.
Calculate the efficiency, η (%), the electromagnetic torque, T_d, and the mechanical and magnetic losses, $p_{(\mathrm{mec+Fe})}$;

b) Assume the switch is closed. What are the new values of current drawn and speed when the electromagnetic torque doubles? State the assumptions made;

c) Sketch qualitatively the shaft torque characteristic of this motor. Explain what happens to it when the supply voltage is varied. Justify.

Results:

a) 83.8 % ; 29.4 Nm ; 370.6 W **b)** 22.4 A ; 888 rpm

Exercise 17

A shunt motor with the following nameplate ratings: 45 kW, 500 V, 100 A, 1600 rpm, has an armature resistance of 0.33 Ω and will use reverse current braking (plugging), from its rated operating condition.

Knowing that the initial braking current must be limited to 150 A, calculate:

a) The useful (shaft) torque, T_u, and the back electromotive force, E', at full load;

b) The value of the braking resistance;

c) Analyse qualitatively the torque developed at the instant the motor stops. What happens from that instant onwards?

d) Consider dynamic (rheostatic) braking. Calculate the initial braking current using the braking resistance found above, starting from the rated operating condition.

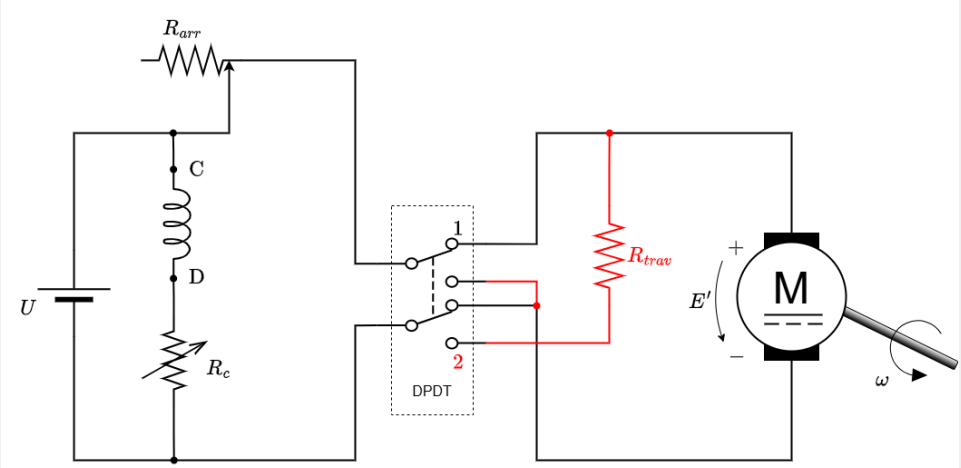
Worked solution:

a) The shaft torque is obtained from the nameplate ratings:

$$T_u = \frac{P_u}{\frac{2\,\pi\,n_n}{60}} = \frac{45\,000}{\frac{2\,\pi\,1600}{60}} = 268.6 \text{ Nm}$$

Calculation of E': $\quad E' = U - R_a\,I_n = 500 - 0.33 \times 100 = 467 \text{ V}$

b) The following figure shows the circuit diagram of the shunt motor fitted with a DPDT (Double Pole, Double Throw) switch for the **reverse current braking (plugging)** manoeuvre:



When the DPDT switches to perform reverse current braking (position 2), the negative armature terminal is connected to the positive terminal of the DC supply, while the positive motor terminal is connected to the negative terminal of the supply through a braking resistance, R_{brk}.

Thus, going round the power loop, the following equation is obtained:

$$-U - E' + (R_a + R_{\text{brk}})\,I_{\text{brk}} = 0 \quad \Rightarrow \quad I_{\text{brk}} = \frac{U + E'}{R_a + R_{\text{brk}}}$$

Note that the braking current, I_{brk}, flows in the opposite direction to the motor current with the DPDT in position 1. Therefore, the torque developed during braking opposes the rotor motion, forcing the speed, ω, to decrease.

As the speed, ω, decreases, the back electromotive force, E', decreases and consequently the current and the torque during the DC motor braking operation also decrease.

Rearranging the previous expression to obtain the braking resistance, R_{brk}, gives:

$$R_{\text{brk}} = \frac{U + E'}{I_{\text{brk}}} - R_a = \frac{500 + 467}{150} - 0.33 = 6.12\ \Omega$$

c) Relating the developed torque, T_d, to the speed, ω, we obtain successively:

$$T_d = K\,\phi\,I_a$$

Particularising for the torque developed during braking, $T_{d_{\text{brk}}}$, considering I_{brk}:

$$T_{d_{\text{brk}}} = K\,\phi\left(\frac{U + E'}{R_a + R_{\text{brk}}}\right) \qquad \text{since:} \qquad E' = K\,\phi\,\omega$$

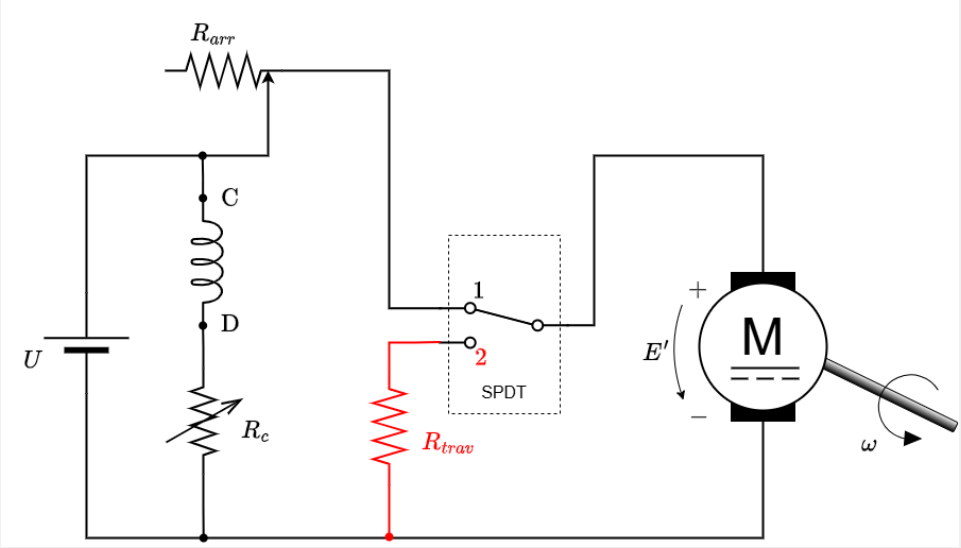
$$T_{d_{\text{brk}}} = \frac{K\,\phi\,U}{R_a + R_{\text{brk}}} + \frac{(K\,\phi)^2\,\omega}{R_a + R_{\text{brk}}} \quad \text{, when:} \quad \omega = 0 \;\Rightarrow\; T_{d_{\text{brk}}} = \frac{K\,\phi\,U}{R_a + R_{\text{brk}}}$$

Analysing the previous expression, the second term decreases during the braking manoeuvre until it becomes zero when: $\omega = 0$.

However, the first term remains constant. Therefore, immediately after the speed reaches zero, if no action is taken, the motor starts will start rotating in the opposite direction. An automatic control must therefore disconnect the motor from the DC supply as soon as its speed reaches zero.

d) Dynamic (or rheostatic) braking consists of disconnecting the motor armature circuit from the DC supply and connecting it across a braking resistance. The DC machine will now operate as a separately excited generator (constant flux) supplying the braking resistance. Since there is no prime mover driving the “generator”, the speed falls sharply within a few moments.

The electrical diagram in the following figure uses an SPDT (*Single Pole, Double Throw*) switch for the dynamic braking manoeuvre:

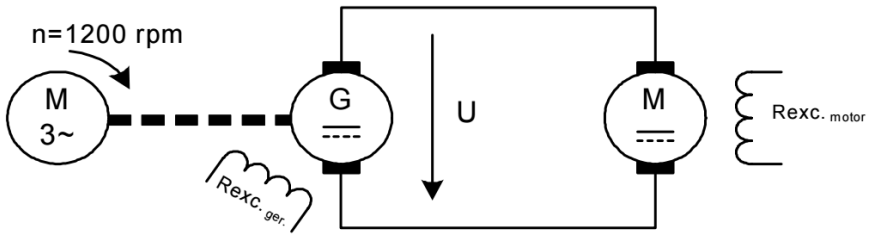


Analysing the flow of the braking current (with the SPDT switch in position 2), at the start of the braking operation, we obtain:

$$I_{\text{brk}} = \frac{E'}{R_a + R_{\text{brk}}} = \frac{467}{0.33 + 6.12} = 72.4\ \text{A}$$

Exercise 18

A separately excited DC motor, 10 kW, 250 V, with an efficiency of 88 %, is supplied by a Ward Leonard system, shown in the following schematic diagram of its operating principle:



The magnetisation characteristics of the generator and the motor, obtained at 1200 rpm, are known:

Generator:	i(A)	0.0	0.4	0.8	1.2	1.6	2.0	$R_a = 0.8\,\Omega$	$R_d = 120\,\Omega$
	E(V)	12	110	200	260	290	310	$U_{\text{exc}} = 240$ V	$\Delta E = 0$ V

Motor:	i(A)	0.0	0.4	0.8	1.2	1.6	$R_a = 1.6\,\Omega$	$R_d = 150\,\Omega$
	E(V)	10	105	190	240	265	$U_{\text{exc}} = 240$ V	$\Delta E = 0$ V

a) Calculate the motor speed and the electromagnetic torque at full load, with: $R_f^{\text{gen}} = 180\,\Omega$ and $R_f^{\text{mot}} = 50\,\Omega$;

b) Using the Ward Leonard system, adjust the motor speed to 1100 rpm at half load. Calculate the value of the field rheostat for this condition;

c) When the generator field current reaches its maximum value, how can the speed of the DC motor be increased? Justify;

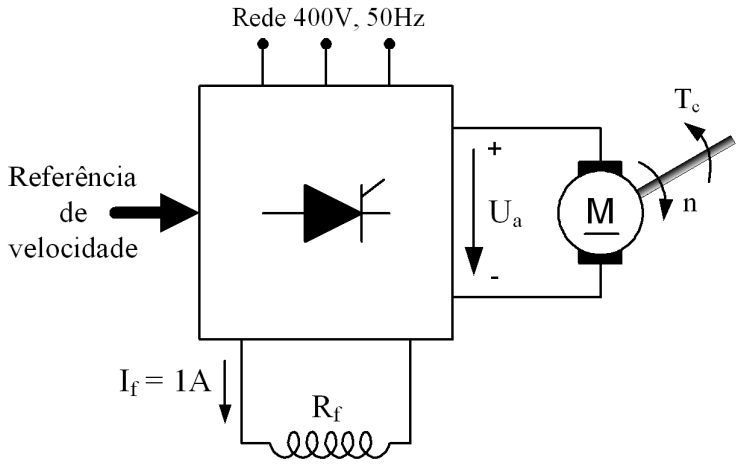
d) State the advantages of replacing the Ward Leonard system with an electronic variable-speed drive.

Results:

a) 455 rpm ; 87 Nm **b)** 53 Ω

Exercise 19

Consider a DC motor supplied from an electronic Ward-Leonard system, as shown in the following figure:



$P_n = 5$ kW

$T_n = 30$ Nm

$I_n = 40$ A

$R_a = 0.8\,\Omega$

The magnetisation characteristic of the machine, obtained at 2000 rpm, is also known:

I_f (A)	0	0.25	0.50	0.75	1.00	1.25	1.50
E_0 (V)	20	160	210	235	250	260	265

a) Determine the voltage applied to the motor armature under rated conditions;

b) Consider the motor operating under the following conditions: $U_a = 195$ V; $n = 1400$ rpm. Now suppose that, using the “Speed reference” command of the electronic variable-speed drive, the armature voltage is suddenly changed to 155 V. Calculate the machine current at the immediately after the change. What has happened? Justify.

Worked solution:

a) Using the rated values provided, the rated speed is obtained from:

$\omega = \dfrac{P_n}{T_n}$ combined with: $\omega = \dfrac{2\,\pi\,n}{60}$, we have:

$$n = \frac{P_n}{T_n}\,\frac{60}{2\,\pi} = \frac{5000}{30}\,\frac{60}{2\,\pi} = 1592 \text{ rpm}$$

On the other hand, for $I_f = 1$ A $\Rightarrow\ K\,\phi = \dfrac{250}{2000}$ V/rpm

Using the speed equation: $n = \dfrac{U_a - R_a\,I_n}{K\,\phi}$, with: $\Delta E = 0$, we obtain:

$$1592 = \frac{U_a - 0.8 \times 40}{250/2000} \ \Rightarrow\ U_a = 231 \text{ V}$$

b) For the specified conditions: $U_a = 195$ V; $n = 1400$ rpm, the motor operating current, I, is:

$$n = \frac{U_a - R_a\, I}{K\,\phi} \;\Leftrightarrow\; 1400 = \frac{195 - 0.8\, I}{250/2000} \;\Rightarrow I = 25 \text{ A}$$

When the supply voltage changes suddenly, $(U_a = 195 \text{ V} \rightarrow 155 \text{ V})$, the speed still remains the same (slow dynamics of a mechanical system *vs.* an electrical system), and the current changes to a new value, given by I':

$$n = \frac{U_a - R_a\, I'}{K\,\phi} \;\Leftrightarrow\; 1400 = \frac{155 - 0.8\, I'}{250/2000} \;\Rightarrow I' = -25 \text{ A}$$

With the sudden change in voltage, U_a, the back electromotive force becomes $E' > U_a$, reversing the direction of the current. In other words, a braking process has started, in which the current, I', will follow the evolution of E', being restored to its previous value, $(I = 25 \text{ A})$, once the speed stabilises.

If the electronic variable-speed drive supports bidirectional current, **regenerative braking** occurs: the current is returned to the electrical supply during the braking as long as $E' > U_a$.

Bibliography

– [Guru & Hiziroğlu, 2003]: Bhag S. Guru, Hüseyin R. Hiziroğlu, Electric Machinery and Transformers, 3rd Ed., Oxford University Press, 2003.